

Contents

$$V_s = V_p \times \left(\frac{N_s}{N_p} \right) = 240 \times \left(\frac{50}{200} \right) = 60 \text{ V}$$

V_s	Secondary Voltage (Output Voltage, in Volts)
V_p	Primary Voltage (Input Voltage, in Volts)
N_s	Number of turns in the Secondary winding
N_p	Number of turns in the Primary winding

$$V_{\text{rms}} = V_{\text{peak}} \times 0.707 = 170 \times 0.707 \approx 120 \text{ V}$$

V_{rms}	Root Mean Square Voltage (Effective AC Voltage)
V_{peak}	Peak Voltage (Maximum amplitude of the waveform)
0.707	The constant representing $\frac{1}{\sqrt{2}}$ for sinusoidal waves

$$V_{\text{rms}} = V_{\text{peak}} \times 0.707 = 170 \times 0.707 \approx 120 \text{ V}$$

V_{rms}	Root Mean Square Voltage (Effective AC Voltage)
V_{peak}	Peak Voltage (Maximum amplitude of the waveform)
0.707	The constant representing $\frac{1}{\sqrt{2}}$ for sinusoidal waves

$$X_c = \frac{1}{2\pi fC}$$

X_c	Capacitive Reactance (Opposition to AC current flow, in Ohms Ω)
f	Frequency of the AC signal (in Hertz, Hz)
C	Capacitance of the capacitor (in Farads, F)
π	Mathematical constant Pi (≈ 3.14159)

$$X_L = 2\pi fL$$

X_L	Inductive Reactance (Opposition to AC current flow, in Ohms Ω)
f	Frequency of the AC signal (in Hertz, Hz)
L	Inductance of the inductor (in Henrys, H)
π	Mathematical constant Pi (≈ 3.14159)

$$P = \sqrt{3} \times V \times I \times \text{PF} = 1.732 \times 480 \times 10 \times 1.0 \approx 8,314 \text{ W}$$

P	Total Three-Phase Power (in Watts, W)
$\sqrt{3}$	Line-to-line multiplier constant for 3-phase systems (≈ 1.732)
V	Line-to-Line Voltage (in Volts, V)
I	Line Current (in Amps, A)
PF	Power Factor (a ratio between 0.0 and 1.0, representing efficiency)

$$R_{\text{total}} = R_1 + R_2 + R_3 = 10 + 15 + 25 = 50 \, \Omega$$

R_{total}	Total Resistance of the series circuit (in Ohms, Ω)
R_1, R_2, R_3	Individual resistors connected in series (in Ohms, Ω)

$$P = I^2 \times R = (4)^2 \times 12 = 16 \times 12 = 192 \, \text{W}$$

P	Power dissipated by the resistor (in Watts, W)
I	Current flowing through the resistor (in Amperes, A)
I^2	The current squared ($I \times I$)
R	Resistance value of the component (in Ohms, Ω)

$$P = \frac{V^2}{R} = \frac{(240)^2}{60} = \frac{57,600}{60} = 960 \, \text{W}$$

P	Power dissipated by the circuit (in Watts, W)
V	Voltage applied across the resistor (in Volts, V)
V^2	The voltage squared ($V \times V$)
R	Resistance value of the component (in Ohms, Ω)

$$I_1 = \frac{120}{40} = 3 \, \text{A}, \quad I_2 = \frac{120}{60} = 2 \, \text{A}$$

$$I_{\text{total}} = I_1 + I_2 = 3 + 2 = 5 \, \text{A}$$

I_{total}	Total current entering or leaving the parallel circuit (in Amperes, A)
I_1, I_2	Individual currents flowing through each parallel branch (in Amperes, A)
KCL	Kirchhoff's Current Law (Sum of currents entering a node = sum of currents leaving)

Kirchhoff's Voltage Law (KVL) states that around any closed loop in a circuit,
The sum of all voltage drops equals the source voltage

- KVL: The algebraic sum of all voltage drops around any closed loop equals the applied source voltage — General Theory (Kirchhoff's Laws).

Kirchhoff's Current Law (KCL) states that at any node in a circuit, **The sum of currents entering the node equals the sum of currents leaving**

$$V_p \times I_p = V_s \times I_s \implies I_s = \frac{V_p \times I_p}{V_s} = \frac{120 \times 8}{480} = \frac{960}{480} = 2 \text{ A}$$

V_p	Primary Voltage (Input Voltage, in Volts)
I_p	Primary Current (Input Current, in Amperes)
V_s	Secondary Voltage (Output Voltage, in Volts)
I_s	Secondary Current (Output Current, in Amperes)
$\implies$	LaTeX symbol for "implies" or "therefore"

$$Q = \sqrt{S^2 - P^2} = \sqrt{10,000^2 - 8,000^2} = \sqrt{100,000,000 - 64,000,000} = \sqrt{36,000,000} = 6,000 \text{ VAR}$$

Q	Reactive Power (measured in Volt-Amperes Reactive, VAR)
S	Apparent Power (total power delivered to the circuit, measured in Volt-Amperes, VA)
P	Real / Active Power (actual work performing power, measured in Watts, W)
S^2, P^2	The respective power values squared

$$V_s = V_p \times \left(\frac{N_s}{N_p} \right) = 240 \times \left(\frac{50}{200} \right) = 60 \text{ V}$$

V_s	Secondary Voltage (Output Voltage, in Volts)
V_p	Primary Voltage (Input Voltage, in Volts)
N_s	Number of turns in the Secondary winding
N_p	Number of turns in the Primary winding

$$V_{\text{rms}} = V_{\text{peak}} \times 0.707 = 170 \times 0.707 \approx 120 \text{ V}$$

V_{rms}	Root Mean Square Voltage (Effective AC Voltage)
V_{peak}	Peak Voltage (Maximum amplitude of the waveform)
0.707	The constant representing $\frac{1}{\sqrt{2}}$ for sinusoidal waves

$$V_{\text{rms}} = V_{\text{peak}} \times 0.707 = 170 \times 0.707 \approx 120 \text{ V}$$

V_{rms}	Root Mean Square Voltage (Effective AC Voltage)
V_{peak}	Peak Voltage (Maximum amplitude of the waveform)
0.707	The constant representing $\frac{1}{\sqrt{2}}$ for sinusoidal waves

$$X_c = \frac{1}{2\pi fC}$$

X_c Capacitive Reactance (Opposition to AC current flow, in Ohms Ω)
 f Frequency of the AC signal (in Hertz, Hz)
 C Capacitance of the capacitor (in Farads, F)
 π Mathematical constant Pi (≈ 3.14159)

$$X_L = 2\pi fL$$

X_L Inductive Reactance (Opposition to AC current flow, in Ohms Ω)
 f Frequency of the AC signal (in Hertz, Hz)
 L Inductance of the inductor (in Henrys, H)
 π Mathematical constant Pi (≈ 3.14159)

$$P = \sqrt{3} \times V \times I \times \text{PF} = 1.732 \times 480 \times 10 \times 1.0 \approx 8,314 \text{ W}$$

P Total Three-Phase Power (in Watts, W)
 $\sqrt{3}$ Line-to-line multiplier constant for 3-phase systems (≈ 1.732)
 V Line-to-Line Voltage (in Volts, V)
 I Line Current (in Amps, A)
 PF Power Factor (a ratio between 0.0 and 1.0, representing efficiency)

$$R_{\text{total}} = R_1 + R_2 + R_3 = 10 + 15 + 25 = 50 \Omega$$

R_{total} Total Resistance of the series circuit (in Ohms, Ω)
 R_1, R_2, R_3 Individual resistors connected in series (in Ohms, Ω)

$$P = I^2 \times R = (4)^2 \times 12 = 16 \times 12 = 192 \text{ W}$$

P Power dissipated by the resistor (in Watts, W)
 I Current flowing through the resistor (in Amperes, A)
 I^2 The current squared ($I \times I$)
 R Resistance value of the component (in Ohms, Ω)

$$P = \frac{V^2}{R} = \frac{(240)^2}{60} = \frac{57,600}{60} = 960 \text{ W}$$

P	Power dissipated by the circuit (in Watts, W)
V	Voltage applied across the resistor (in Volts, V)
V^2	The voltage squared ($V \times V$)
R	Resistance value of the component (in Ohms, Ω)

$$I_1 = \frac{120}{40} = 3 \text{ A}, \quad I_2 = \frac{120}{60} = 2 \text{ A}$$

$$I_{\text{total}} = I_1 + I_2 = 3 + 2 = 5 \text{ A}$$

I_{total}	Total current entering or leaving the parallel circuit (in Amperes, A)
I_1, I_2	Individual currents flowing through each parallel branch (in Amperes, A)
KCL	Kirchhoff's Current Law (Sum of currents entering a node = sum of currents leaving)

Kirchhoff's Voltage Law (KVL) states that around any closed loop in a circuit, **The sum of all voltage drops equals the source voltage**

- KVL: The algebraic sum of all voltage drops around any closed loop equals the applied source voltage — General Theory (Kirchhoff's Laws).

Kirchhoff's Current Law (KCL) states that at any node in a circuit, **The sum of currents entering the node equals the sum of currents leaving**

$$V_p \times I_p = V_s \times I_s \implies I_s = \frac{V_p \times I_p}{V_s} = \frac{120 \times 8}{480} = \frac{960}{480} = 2 \text{ A}$$

V_p	Primary Voltage (Input Voltage, in Volts)
I_p	Primary Current (Input Current, in Amperes)
V_s	Secondary Voltage (Output Voltage, in Volts)
I_s	Secondary Current (Output Current, in Amperes)
$\implies$	LaTeX symbol for "implies" or "therefore"

$$Q = \sqrt{S^2 - P^2} = \sqrt{10,000^2 - 8,000^2} = \sqrt{100,000,000 - 64,000,000} = \sqrt{36,000,000} = 6,000 \text{ VAR}$$

Q	Reactive Power (measured in Volt-Amperes Reactive, VAR)
S	Apparent Power (total power delivered to the circuit, measured in Volt-Amperes, VA)
P	Real / Active Power (actual work performing power, measured in Watts, W)
S^2, P^2	The respective power values squared

An inductive load causes the current waveform to do which of the following relative to the voltage waveform to **lag voltage**